

Radar Search Pipeline TO BE 0.7.6.4.18.1

Review design for a planned Radar search pipeline change

Status: Implemented design input

Product area: Radar candidate discovery pipeline

Slice: 0.7.6.4.18.1 Candidate discovery and signal monitoring runtime split

Last updated: 2026-07-06

Generated PDF: docs/radar/to-be/RADAR_SEARCH_PIPELINE_TO_BE_0.7.6.4.18.1.pdf

Goal

Candidate discovery becomes a producer/handoff pipeline. It finds and qualifies candidate entities, records checkpoints, source obligations, budgets, coverage, and candidate universe diagnostics, then projects signal-monitoring intent as pending work. It no longer owns recurring signal evaluation as the normal runtime path.

Intended Runtime Split

Normal candidate discovery:

1. Planning and plan acceptance remain unchanged.
2. Discovery, retrieval, extraction, recovery, registry/entity resolution, checkpoints, coverage, expansion, and disambiguation remain unchanged.
3. The pre-signal checkpoint remains mandatory.
4. If the checkpoint continues, candidate discovery does not call providers for `signal_search`.

5. It projects one handoff status per configured signal task and candidate scope row:

- `search_status="not_searched_pending_signal_monitoring";`
- `not_searched_reason="pending_signal_monitoring";`
- `signal_task_count=0.`

6. Final metadata exposes:

- `signal_execution_mode;`
- `signal_monitoring_handoff_status;`
- `signal_monitoring_pending_count.`

Explicit compatibility mode:

- `signal_execution_mode="inline_compatibility"` preserves the old embedded `signal_search` execution for legacy tests and hidden callers.
- This mode may produce searched and searched-negative signal states because provider work actually ran.
- It is not the normal product runtime.

Behavior Rules

- Unknown `signal_execution_mode` values fall back to handoff.
- `not_observed` is legal only when a signal was actually searched by signal monitoring or explicit inline compatibility.
- A blocked pre-signal checkpoint still produces existing `not_searched_policy_limited` / `stopped-for-review` statuses.
- Candidate discovery may keep signal task definitions in the execution plan as handoff intent, but it must not spend signal-monitoring budget.
- Candidate discovery must not import the `radar.signal_monitoring` runtime package.

Validation Plan

- Focused tests prove default candidate discovery does not call provider stage `signal_search`.
- Compatibility tests explicitly opt into `inline_compatibility`.
- Adaptive/live/API regression tests prove candidate/source/checkpoint/dossier shape remains stable.
- Signal-monitoring recorded tests prove the separate pipeline still runs.
- AS IS Markdown/PDF is regenerated after implementation.